

DAISY ZHOU

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EDUCATION

Columbia University

New York, NY

MS in Operations Research, GPA 3.83

Aug 2025 - Dec 2026

- Coursework: Dynamic Pricing & Revenue Management, Transportation Analytics & Logistics, Machine Learning, Applied Data Science, Stochastic Models, Optimization (Linear, Integer, Nonlinear, Network, Combinatorial)

Georgia Institute of Technology

Atlanta, GA

BS in Industrial Engineering, GPA 4.0

Dec 2022

WORK EXPERIENCE

The Boeing Company

Everett, WA

Systems Engineer

Feb 2023 - Jun 2025

- Modeled SysML functional and logical architecture for fuselage assembly and 737 wing panel assembly manufacturing lines in Cameo to consolidate traditional document-based materials into a digitalized single source of truth
- Co-authored 4 user guides on building logical and functional architecture, state machines, and manufacturing bill of materials in Cameo to drive data consistency and uniform standards across all Production System use cases
- Led pathfinding efforts with domain experts to determine best modeling practices on Production System design and development, advancing Model Based Systems Engineering maturity and department-wide adoption
- Developed and implemented custom data queries in Cameo to build standardized table templates that optimize data reporting processes and replace error-prone Excel artifacts

Piedmont Atlanta Hospital

Atlanta, GA

Simulation Consultant Intern

Sep 2021 - May 2022

- Cleaning and standardized data in Excel, performing data imputation across 30+ surgeon preference cards using well-defined modeling assumptions for model accuracy and validity
- Modeled instrument storage queues, surgical cart batching, and off-nominal scenarios (surgery rescheduling, missing instrument trays, and failed quality audits) through logic-based process flows in Simio
- Computed optimal instrument tray and staffing levels in OptQuest, achieving 99% fill rates at 80% operating capacity level + 20% buffer with projected annual savings of over \$3 million

Delta Air Lines

Atlanta, GA

Reservations Technology and Data Analytics Intern

May 2021 - Aug 2021

- Analyzed customer service staffing supply and demand using annual schedule and workload data to identify capacity constraints and periods exceeding 80% utilization
- Developed Excel VBA reporting tool to visualize staffing capacity levels, reducing report generation time by 95% and supporting long term workforce planning

ACADEMIC & PROJECT EXPERIENCE

Course Assistant: Optimization Models and Methods

Jan 2026 - May 2026, Sep 2026 - Present

- Graded coursework, proctored exams, and supported students with linear, mixed integer, nonlinear, and network optimization concepts

Comparative Survival Modeling for Incident Myocardial Infarction (MI) Prediction

Feb 2026 - May 2026

- Engineered high-dimensional clinical features from ICU vitals, labs, comorbidities, and medications collected during the first 24 hours of admission
- Built Cox PH, Elastic Net Cox, Random Survival Forest, and Gradient Boosting survival models to predict time-to-first-MI among 33k+ ICU patients, achieving average C-index of 0.70

IBM Job Salary Prediction and Labor Market Analytics

April 2026 - May 2026

- Developed end-to-end ML pipeline on large-scale web-scraped IBM job posting data using feature engineering, clustering, and regression modeling
- Built Gradient Boosting, Random Forest, and Ridge Regression models with cross-validation and hyperparameter tuning, achieving $R^2 = 0.85$ for salary prediction

Dynamic Pricing and Revenue Optimization

April 2026 - May 2026

- Formulated multi-phase stochastic pricing strategy using Bayesian demand learning, Thompson Sampling, Gradient Boosting for revenue optimization in nonstationary market conditions
- Designed competitor-aware pricing policies using contextual features and adaptive demand estimation within a dynamic competitive simulation

Columbia Transportation Route Optimization

Nov 2025 - Dec 2025

- Designed dynamic batching and routing optimization model for Columbia's micro-transit shuttle service using pickup-and-delivery Traveling Salesman Problem, reducing average passenger wait time by 43%
- Validated model performance through simulation and sensitivity analysis using historical ride requests and OpenStreetMap road network data

TECHNICAL SKILLS

Programming: Python (NumPy, pandas), R, SQL, VBA

Optimization & Simulation: Gurobi, OPL/CPLEX, DOcplex, Simio, OptQuest, SciPy

Machine Learning & Statistics: scikit-learn, statsmodels, scikit-survival, lifelines

BI & Platforms: Excel, Tableau, BigQuery, Google Cloud Platform